

5.3 Amazon Q Developer Getting Started

Lesson 2 of 9

Introduction to Amazon Q Developer

Lesson objectives

In this lesson, you will learn:

- How Amazon Q Developer works.
- What problems Amazon Q Developer solves.
- The benefits of Amazon Q Developer.
- Amazon Q Developer pricing.



What does Amazon Q Developer do?

Amazon Q Developer is a generative AI powered assistant. It helps you understand, build, extend, and operate Amazon Web Services (AWS) applications throughout the software development lifecycle. It uses natural language processing to answer questions on AWS architecture, best practices, documentation, and support, providing contextually relevant and actionable answers.

Amazon Q Developer is accessible through multiple channels such as the AWS Management Console, where it can analyze and troubleshoot issues across services. In addition, it is available from:

- The AWS website
- AWS documentation pages
- The AWS mobile app
- IDEs with the Amazon Q extension
- The AWS Chatbot for Microsoft Teams and Slack
- Amazon CodeCatalyst

What problems does Amazon Q Developer solve?

Amazon Q Developer reduces the time spent on manual tasks, such as coding, testing, upgrading, troubleshooting, and optimizing your code. Amazon Q Developer is available wherever you need it: in your IDE, AWS console, terminal, or Slack. Amazon Q Developer is your developer companion.

To recognize problems that Amazon Q Developer can solve, review the following information.



Increase productivity

Amazon Q Developer streamlines repetitive tasks and accelerates the development workflow. It provides software development assistance, including code explanation, code generation, and code improvements such as debugging and optimization.



Onboard developers new to AWS

Amazon Q Developer helps new developers understand AWS services and how to select the right services for their needs. It provides guidance on how to start their journey in AWS and recommendations based on the AWS Well-Architected Framework.



Develop code features

You can develop code features and projects in your programming language of choice. You explain the feature you want to develop. Then, Amazon Q Developer uses the context of your current project to generate a detailed implementation plan that includes the code for the changes you described.



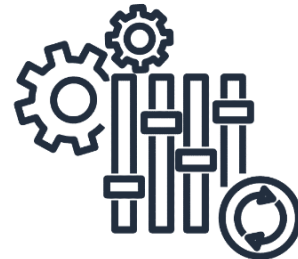
Transform code

Within IDEs, Amazon Q Developer can update the language version of your code files. Currently, Amazon Q Code Transformation supports updating Java 8 and Java 11 code to Java 17. For more information, see [Upgrade Language Versions with the Amazon Q Developer Agent for Code Transformation](#).



Troubleshoot issues

You can often run into errors or issues when deploying and operating applications. Amazon Q Developer helps troubleshoot and resolve common deployment, configuration, and runtime issues through natural language conversations. It uses knowledge from AWS documentation.



Autonomous agents

Amazon Q Developer Agents take a lot of work out of complex, multistep tasks. The agent for software development helps with implementing features, documenting code, and bootstrapping new projects, all from a single prompt.

For more information, see [Develop software with the Amazon Q Developer Agent for software](#)

What are the benefits of Amazon Q Developer?

Amazon Q Developer accelerates the software development lifecycle by boosting employee's productivity leading to lower cost to build an application.

To learn more about the benefits of Amazon Q Developer, expand each of the following sections.

Accelerates the entire software development process	+
Boosts employee productivity and accelerates new feature development	+
Saves time and reduces costs across organization	+

Accelerates the entire software development process

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More than 70 percent of developer time is spent on undifferentiated activities slowing down creativity and innovation. These activities include writing boilerplate code, developing and running unit tests, and translating code from one language to another.

Boosts employee productivity and accelerates new feature development

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Amazon Q Developer expedites the development of new features. It can boost productivity by generating new software code suggestions for application development tasks.

Saves time and reduces costs across organization

—

Amazon Q Developer includes security scans that you can run in the IDE. These scans help you find and correct potential vulnerabilities earlier in the application lifecycle, thereby lowering the cost, time, and risk of application development.

Architecture and Use Cases

Lesson objectives

In this lesson, you will learn:

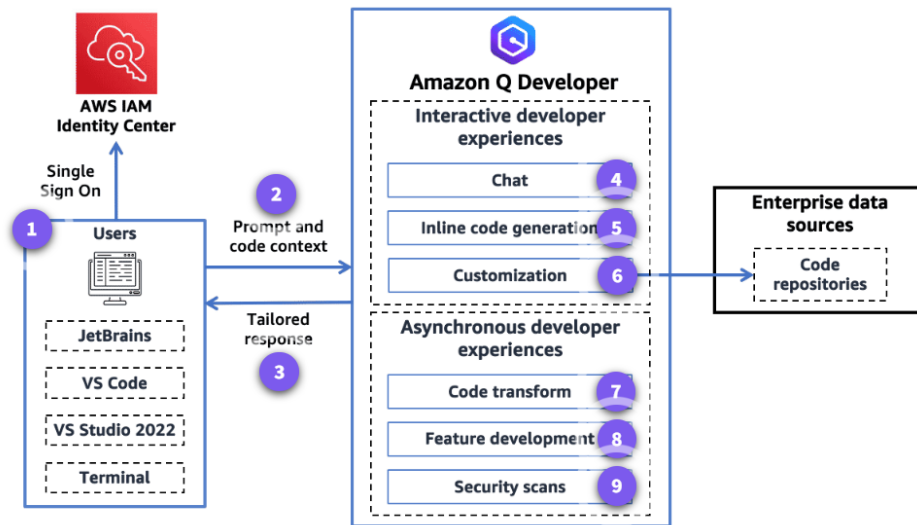
- The technical concepts of Amazon Q Developer.
- Typical use cases for Amazon Q Developer.
- The requirements to implement Amazon Q Developer in a real-world scenario.

How is Amazon Q Developer used to architect a cloud solution?

Amazon Q Developer helps you explore new AWS capabilities, review your resources, analyze your bill, and architect solutions. It's an expert in AWS well-architected patterns, documentation, solutions implementations, and more. The following architecture logically illustrates how you can use Amazon Q Developer.

Please note that in the diagram, use of the AWS IAM Identity Center is only available when using the Amazon Q Developer Pro tier. All of the other features are available regardless of tier.

To learn more about the architecture of Amazon Q Developer, choose each of the numbered markers.



What are the basic technical concepts of Amazon Q Developer?

- Using in AWS Management Console
- Access control
- IDE plugin
- Terminal plugin
- Customization
- Chat

What are typical use cases for Amazon Q Developer?

Amazon Q Developer plays a crucial role in enhancing the efficiency and effectiveness of the software development lifecycle (SDLC). The five phases are plan, create, test and secure, operate, and maintain and modernize.

- **Plan:** Navigating through vast amounts of technical documentation and examples can be daunting and time-consuming. Amazon Q Developer simplifies this task by providing targeted, business-specific guidance and code explanations through conversational coding, aiding in the planning phase of application development.
- **Create:** As an inline coding assistant in both IDEs and the AWS CLI, Amazon Q Developer accelerates the development of new features and the maintenance of existing infrastructure. Its conversational coding capabilities help developers swiftly implement and iterate on their ideas.
- **Test and secure:** Ensuring that code changes are robust and secure is crucial. Amazon Q Developer enhances this phase by assisting with generating unit tests and identifying security vulnerabilities. It also offers remediation advice through its integrated security scanning features, helping developers enforce best security practices effortlessly.
- **Operate:** For AWS developers, Amazon Q Developer is indispensable, offering expert troubleshooting and optimization across various AWS services. These services include Amazon S3, AWS Lambda, Amazon Elastic Compute Cloud (Amazon EC2), and Amazon Elastic Kubernetes Service (Amazon EKS). Integrated directly within your IDE, terminal, or console, Amazon Q Developer helps ensure smooth operations and efficient problem resolution.
- **Maintain and modernize:** The Amazon Q Code Transformation feature of Amazon Q Developer streamlines the task of modernizing code and updating dependencies to newer programming languages. This feature reduces complexity and minimizes the risk of errors during the modernization process which helps you keep your applications up to

What else should I keep in mind about Amazon Q Developer?

- Be specific and clear in prompts to get your desired responses. State the task directly and provide details.
 - Clarity involves accuracy and specificity, leaving little room for misinterpretation.
 - Intent refers to the purpose or goal behind the prompt.
 - Context includes the surrounding information or environment related to the prompt.
- Use relevant examples to provide additional context and guide better outputs.
- Use an iterative approach. Refine and rephrase prompts based on responses.
- Break down complex queries into smaller, manageable parts.

Lesson 4 of 9

How Do I Set Up a Development Environment to Use with Amazon Q Developer?

Lesson objectives

In this lesson, you will learn how to install the Amazon Q extension in Visual Studio Code and establish an AWS Builder ID.

Lesson 5 of 9

How Do I Interact with Amazon Q Developer in Visual Studio Code?

Lesson objectives

In this lesson, you will learn how to interact with Amazon Q Developer in Visual Studio Code.

Lesson 6 of 9

How Do I Transform Code from Java 8 to Java 17 by Using Amazon Q Developer?

Lesson objectives

In this lesson, you will learn how to transform code from Java 8 to Java 17 by using Amazon Q Developer.

How Do I Ask Amazon Q Developer to Implement a New Feature to a Project?

Lesson objectives

In this lesson, you will learn how to ask Amazon Q Developer to implement a new feature to a project.